

SHUAI YANG

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EDUCATION

Hong Kong University of Science and Technology (HKUST) 2026 – 2030 (expected)

- **Ph.D. Student**, Department of Computer Science and Engineering & Cheng Kar-Shun Robotics Institute.
- **Supervisor**: Yinghao Xu (HKUST).

Zhejiang University (ZJU), Hangzhou, China 2021 – 2026

- **Bachelor's Degree in Information Engineering** (Shannon Excellence Class)
- **Elite Program**: Advanced Honor Class of Engineering Education, Chu Kochen Honors College
- **Academic Performance**: Major GPA: 92.20, Rank: 5/128 (Top 5%)

Research Interests: Video-Action Models, Vision-Language-Action Models, Real-World Data Scaling, Agentic Robotic Systems.

EXPERIENCE

(MANIPULATION: 🤖 MULTIMODAL: 🗨️)

1. Native Video-Action Pretraining for Generalizable Robot Control 🤖 (arXiv) 2026

Qihang Zhang, Lin Li, Luyao Zhang, **Shuai Yang**, et al.

Core Contributor

- Presenting **LingBot-VA 2.0**, a natively pretrained video-action foundation model combining semantic visual-action tokenization, causal pretraining, sparse mixture-of-experts modeling, and asynchronous inference for generalizable robot control.

2. From Foundation to Application: Improving VLA Models in Practice 🤖 (arXiv) 2026

- Advancing **LingBot-VLA 2.0** through large-scale robot and egocentric data pretraining, expanded whole-body action spaces, and predictive dynamics modeling for cross-embodiment mobile manipulation.

3. Causal World Modeling for Robot Control 🤖 (RSS 2026) 2026

Lin Li, Qihang Zhang, Yiming Luo, **Shuai Yang**, et al.

- Introducing **LingBot-VA**, an autoregressive diffusion framework that jointly models future video and actions, with closed-loop rollout and asynchronous inference for long-horizon robot manipulation.

4. A Pragmatic VLA Foundation Model 🤖 (arXiv) 2026

Wei Wu*, Fan Lu*, Yunnan Wang*, **Shuai Yang***, Shi Liu*, Fangjing Wang*, et al.

- Developing **LingBot-VLA**, a foundation model pretrained on 20,000 hours of real-world data from nine dual-arm platforms, supporting cross-task generalization and efficient adaptation to new robot embodiments.

5. Vision-Language-Action Instruction Tuning: From Understanding to Manipulation 🤖 2025

- Presenting the first VLA instruction tuning dataset and bench to facilitate language-steered manipulation.
- Introducing **InstructVLA**, which is capable of interactive manipulation with language reasoning abilities.

6. GenManip: A Simulator for Generalizable Tabletop Manipulation 🤖 (Accepted by CVPR) 2025

Gao Ning*, Yilun Chen*, **Shuai Yang***

- Building the first reasoning-focused tabletop manipulation benchmark for embodied agents in IsaacSim, supported by a large-scale collection of well-annotated 3D assets, including rigid and articulated objects.

7. Transferring Latent Motion Across Time for Multi-Frame Prediction in Manipulation 🤖 2025

Hao Li*, **Shuai Yang***

- Developing the first VLA model that utilizes temporal information for enhanced manipulation.

8. Grounded 3D-LLM with Referent Tokens 🗨️ 2024

Yilun Chen*, **Shuai Yang***, Haifeng Huang*

- Presenting the first large-scale phrase level 3D scene-language correspondence dataset, facilitating contrastive language-scene pre-training and harmonizing inputs into a unified embedding space.
- Introducing the first generalist LLM for Grounded 3D Scene understanding.

9. MMScan 🗨️ (Accepted by NeurIPS 2024) 2023–2024

Ruiyuan Lyu*, Tai Wang*, Jingli Lin*, **Shuai Yang***

- Building the first largest multimodal 3D scene dataset and benchmark with hierarchical annotations.

10. PointLLM V2 (Accepted by TPAMI 2025) 🗨️ 2024

Runsun Xu*, **Shuai Yang***

- Enhancing the overall captioning and dialogue performance of PointLLM V1 while equipping it with part-referencing capabilities and embodied understanding.

Service

Reviewer: NeurIPS, ICLR

INTERSHIPS

- Research Intern, RobbyAnt (2025–present). Research on vision-language-action (VLA) and video-action (VA) model pretraining, real-world robotic deployment, scaling robot learning datasets, and the design of novel robotic embodiments.
- Research intern at ShangHai AI Lab -Open Robot Lab. (2024-2025)
- Research intern at CMU Safety AI Lab, focusing on developing safe autonomous driving agents. (2024)

HONORS & AWARDS

- 2021-2022 Academic Year First-Class Scholarship, Zhejiang University (3%)
- 2023 Interdisciplinary Contest In Modeling Meritorious Winner (7%)
- 2022-2023 Academic Year Samsung Scholarship (5%)
- 2025 Excellent Bachelor's Thesis